



Task 3 — Develop and Track Your Prototype Progress

Project: AgriSmart — IoT Smart Irrigation for Egyptian Farmers

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MVP Goal

Deliver a working prototype that measures soil moisture, temperature, and salinity/EC at the root zone using an ESP32-based field unit, transmits that data to the AgriSmart backend, and surfaces it to the farmer through an Arabic-first mobile application — with the option to remotely open/close an irrigation valve from the app. The MVP is deliberately scoped to a single sensing/control zone rather than a multi-zone or multi-farm deployment, consistent with the system design validated in Task 2.

What Has Been Implemented

Backend (complete and tested)

- Device authentication, independent from user/farmer login, so each ESP32 unit is verified by its own credentials before any telemetry or command is accepted.
- Telemetry ingestion and schema validation for soil moisture, temperature, and EC readings.
- Irrigation command state machine (queued → sent → acknowledged → executing → completed/failed), so a command being sent is never assumed to mean the valve actually moved.
- Farm-scoped authorization (anti-IDOR) ensuring a farmer can only see and control their own farm's devices and valves.
- Full backend test suite passing, and lint/dependency-audit checks clean.

Mobile application (design stage)

- UI/UX wireframes completed for the Arabic-first interface: live sensor readings, historical data view, and the irrigation valve control screen.

***Note for the team:** The wireframes exist but the app has no functional code connected to the backend API yet — this is explicitly in progress, not complete. See Next Sprint below.*

Hardware (assembled, firmware pending)

- All core components (ESP32, capacitive soil moisture probe, digital temperature probe, EC probe, relay, and solenoid valve) have been procured and physically assembled/wired together on the bench.

***Note for the team:** The hardware is wired but not yet running its own firmware — the ESP32 does not yet read the sensors or talk to the backend. This is the main item being closed out this week; see Next Sprint below.*

Hardware Components Status

Component	Procurement Status	Build/Integration Status
ESP32 microcontroller module	Procured & assembled	Wired to sensors and relay; firmware not yet loaded
Capacitive soil moisture probe	Procured & assembled	Physically connected to ESP32 analog input; reading routine not yet coded
Waterproof digital soil temperature probe	Procured & assembled	Physically connected; reading routine not yet coded
Soil EC / salinity probe	Procured & assembled	Physically connected; reading routine not yet coded
Relay + solenoid irrigation valve	Procured & assembled	Wired to ESP32 relay output; command-trigger firmware not yet coded
Power supply (bench power for current testing)	In place for bench testing	Solar/battery field-power path not yet built — bench-only at this stage
Field enclosure (IP65/IP67)	Not yet sourced	Planned after the electronics are confirmed working on the bench

Evidence of Work

The following evidence supports the status reported above and should be attached to this document before submission:

- Screenshot(s) of the backend automated test suite passing (terminal output showing test counts and results).
- Screenshot(s) of the Arabic-first mobile app wireframes.
- Photo(s) of the physically assembled ESP32 + sensor + relay bench setup.
- Optional: a short (≤ 60 second) video of the wiring, once firmware is loaded and the first sensor reading is captured.

Note for the team: No evidence files have been attached to this draft yet — placeholders only. Please send the actual screenshots/photos so they can be inserted before submission; nothing should be represented as demonstrated unless it is actually captured.

Challenges Encountered

- Coordinating hardware assembly (multiple analog/digital sensors sharing one ESP32's limited GPIO/ADC pins) with the backend's already-defined telemetry schema, to make sure firmware output matches what the API expects without rework.



- Sequencing the mobile app build so that it consumes the already-authenticated device/telemetry API correctly on the first attempt, rather than needing a second integration pass.
- Compressed timeline: hardware, firmware, and app functional development are converging in the same week ahead of the deadline.

Note for the team: *These are the challenges as currently understood by the team; edit or add to this list with anything specific that came up during actual assembly/coding, so the report reflects what genuinely happened rather than a generic list.*

Next Sprint Plan

Target: complete the remaining integration work within this week, ahead of the next checkpoint.

- Write and load ESP32 firmware to read all three sensors and publish telemetry to the backend's existing ingestion endpoint.
- Implement firmware handling of the irrigation command flow (receive command → drive relay → report acknowledgment/execution result back to the backend), matching the state machine already built server-side.
- Build the functional mobile app screens on top of the completed wireframes, wired to the live backend API (live readings, history view, valve control).
- Run a first end-to-end integration test: real sensor reading → backend → app display, and app-issued command → backend → relay → valve.
- Capture real evidence (screenshots of live data flowing, a short demo video) once the end-to-end test succeeds, to attach to the next progress report.

This report reflects actual project status as of the date below: backend complete and tested; hardware procured and assembled with firmware pending; mobile app wireframed with functional development pending. No completed hardware/app functionality is claimed beyond what is stated here.